

Monitoring of Storage Unit in the Grid Structure

SITE : \$.SiteId
 OPERATOR : \$.SiteOperator
 MONITORING PERIOD : \$.MoniPeriod
 EVALUATION DATE : \$.EvaDate

Introduction

This report provides analysis and presentation of measured data for the storage system
 /\$.SiteId

SiteDescription

The storage unit is located at

It enables installation and operation of an enlarged PV-Generator with 2.5 times higher peak power by means of buffering PV generation, ensuring the line power to stay within the limits allowed.

The report is conforming with the EC Guidelines for unified monitoring and performance assesement of storage-units in the grid structure.

Thus, it reveals energy balancing in a subgrid, characterized by both energy surplus and autonomy ratio. Further it quantifies and details contribution of the storage unit to grid control and balancing, and thus relief of the utility interface, resp. power line.

In the form of directed power-flows, monitoring data is suitable for spatial aggregation with results from other storage units in a common grid area.

Further to reporting for storage and distribution system operators, guideline conformant data and monitoring evaluation qualify for sharing performance experience, e.g. in the JRC's storage observatory.

General Data

Storage Unit	Nominal Power/[MW]: /\$.NomStoPow	Nominal Capacity/[MWh]: /\$.NomStoCap
Utility Interface	Nominal Power/[MW]: /\$.NomUIPPow	Site : /\$.SiteId

Energetic Balances

In line with recording format from independent metering, the total energies in the monitoring period sum up to :

	Input		Output	
Subgrid	17.5	MWh	19.5	MWh
Utility	30.0	MWh	30.0	MWh
Storage	28.0	MWh	25.0	MWh

with the total difference between in- and outputs covering the storage losses.

In accumulation of directed powerflows, the energy exchanges of the storage unit with subgrid surpluses(net generation) and deficits(net-load), and with the utility can be distinguished from the direct flows between subgrid and utility.

net-generation to utility	19.58	MWh
storage from net-generation	25.00	MWh
storage from utility	3.21	MWh
storage to utility	2.73	MWh
storage to net-load	25.07	MWh
net-load from utility	30.84	MWh

Exchange Yields

Normalizing the directed powerflows, to the nominal power of the storage unit, is resulting in yields, comparable for units of different sizes. Chart 1 shows the exchange yields in their daily averages .

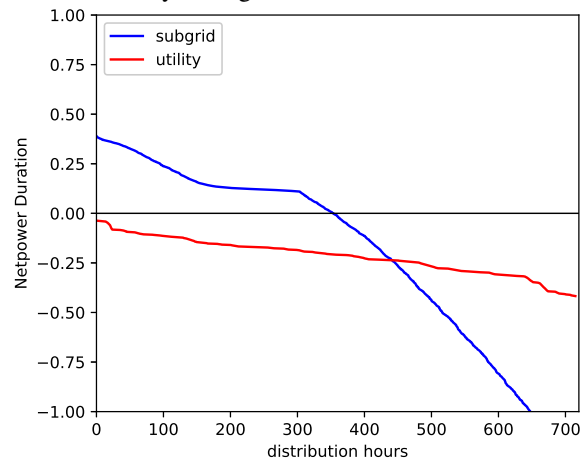


chart 1 : Average Daily Yields